

torch.nn.functional.max_pool2d#

https://pytorch.org/docs/stable/generated/torch.nn.functional.max_pool2d.html

Description:

Applies a 2D max pooling over an input signal composed of several input planes. The order of `ceil_mode` and `return_indices` is different from what seen in `MaxPool2d`, and will change in a future release. See `MaxPool2d` for details. `input` – input tensor (`minibatch, in_channels, iH, iW`) (`\text{minibatch}` , `\text{in\_channels}` , `iH` , `iW`) (`minibatch, in_channels, iH, iW`), `minibatch` dim optional.

Function Signature

```
torch.nn.functional.max_pool2d(input, kernel_size,
stride=None, padding=0, dilation=1, ceil_mode=False,
return_indices=False)[source]#
```

Parameters

Notes & Warnings

NOTE

Note The order of `ceil_mode` and `return_indices` is different from what seen in `MaxPool2d`, and will change in a future release.

Detailed Documentation

Documentation

Applies a 2D max pooling over an input signal composed of several input planes.

The order of `ceil_mode` and `return_indices` is different from what seen in `MaxPool2d`, and will change in a future release.

See `MaxPool2d` for details.

`input` – input tensor (`minibatch, in_channels, iH, iW`) (`\text{minibatch}` , `\text{in\_channels}` , `iH` , `iW`) (`minibatch, in_channels, iH, iW`), `minibatch` dim optional.

`kernel_size` – size of the pooling region. Can be a single number or a tuple (`kH`, `kW`)

`stride` – stride of the pooling operation. Can be a single number or a tuple (`sH`, `sW`).

Default: `kernel_size`

`padding` – Implicit negative infinity padding to be added on both sides, must be ≥ 0 and $\leq \text{kernel_size} / 2$.

`dilation` – The stride between elements within a sliding window, must be > 0 .

`ceil_mode` – If `True`, will use `ceil` instead of `floor` to compute the output shape. This ensures that every element in the input tensor is covered by a sliding window.

`return_indices` – If `True`, will return the `argmax` along with the max values. Useful for `torch.nn.functional.max_unpool2d` later
